

AMD Athlon XP 3000+

Cache-happy

In response to Intel's Hyper-Threaded 3.06 GHz Pentium 4, AMD has released its new XP 3000+ processor, codenamed Barton. The new-born has one significant difference over its predecessor—the Thoroughbred—an additional 256 KB L2 cache. It combines 512 KB of L2 cache with the existing 128 KB of L1 cache, giving you a total of 640 KB on-die cache.

The Barton core is currently available in three speeds—3000+, 2800+ and 2500+.

The Barton requires an FSB of 166 MHz, and the 3000+ is clocked at 2166 MHz—equivalent to the Thoroughbred 2700+. The additional 256 KB L2 cache in Barton adds 300 points to the CPU rating, bringing it to 3000.

We tested the Barton on the MSI KT4 Ultra motherboard that utilises VIA's KT400 chipset. Although the manual claimed that the 3000+ CPU was supported, it required a BIOS upgrade to make it recognise the CPU. Also, the FSB has to be specified as 166 MHz in the BIOS settings. The other components in the test setup included a 256 MB DDR 333 stick, ATi Radeon 9700 PRO accelerator, IBM Deskstar 40 GB hard disk and Samsung CD-RW/DVD combo drive.

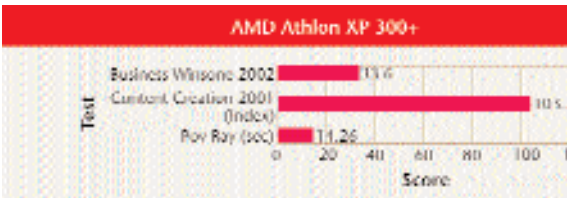
An increase in the on-die cache usually leads to a 10 to 15 per cent increase in heat

dissipation; fortunately, even though Athlon CPUs have been traditionally quite hot, the addition of the cache did not affect the 3000+ much. The thermal grease, heat sink and fan combo perform an excellent job of keeping temperatures down, even while running CPU-intensive tasks for hours.

In the CPU benchmarks, SiSoft Sandra 2003 Pro reported scores of 8,071 MIPS and 3,271 FLOPS in the Arithmetic benchmarks. In the multimedia benchmarks, the scores reported were 11,923 it/sec in the Integer tests and 12,688 it/sec in the FPU SSE tests. PCMark2002 Pro meanwhile, displayed a score of 6,454 marks. Finally, in the CPU-intensive video encoding tests, the CPU performed the operation in a time of 86.58 seconds—a few seconds more than the time taken by the Intel 3.06 GHz CPU.

A P4 3.06 GHz CPU placed in an equivalent system configuration would have produced scores 8 to 10 per cent higher. Even when the 3000+ was replaced by the Thoroughbred 2800+, the scores were almost the same as those of the 3000+—in a few tests, the 3000+ actually scored lower results.

The weakness in graphical applications was reiterated by the 3DMark2001 SE and *Quake III Arena* scores. While 3DMark produced a score of 12,527 marks, the *Quake III* demo ran at 233.2 fps. It scored an impressive 103.3 Winstone units in Content Creation 2001. It also did well in POV-Ray's rendering time for a standard chess-



board scene, clocking a super-fast 14.26 seconds. It is here that the extra cache made its presence felt.

Currently, there is a shortage of Barton CPUs, giving vendors the opportunity to jack up their prices. AMD enthusiasts will proba-

bly have to wait a few weeks for the prices to settle.

SPECIFICATIONS

2166 MHz clock speed
166 MHz FSB, 128 KB L1 cache
512 KB L2 cache, 3000+ Socket A interface

Price: US\$ 580 (approximately Rs 29,000)
Contact: AMD Far East Ltd
Phone: 011-26238620
Fax: 011-2628 364
E-mail: amd@surfgold.com
Web site: www.amd.com

AMD Athlon XP 3000+	
Performance	▶▶▶▶▶
Build Quality	▶▶▶▶▶
Value for money	▶▶▶▶▶
Features	▶▶▶▶▶
OVERALL	▶▶▶▶▶

IndiaAccounts

Online accounting

The latest version of the country's only Web-based accounting software includes modules on Accounts, Inventory and Payroll—a significant improvement over its previous release and somewhat similar to popular accounting packages such as Tally.

IndiaAccounts is capable of the usual tasks of creation of groups, accounts and vouchers, which include all of the basic types such as receipts, payments, sales, purchases and journals.

The major security concerns of loss of data and hacking are taken care of by automatic backup of data on off-site backup servers and 128-bit encryption.

IndiaAccounts can import data directly from Tally. It works on the ASP Model. The

Price: Rs 750 setup charges
Rs 200 monthly maintenance for home users
Rs 500 monthly maintenance for office user
Contact: Microcell Solutions Pvt Ltd
Phone: 022-26206442
E-mail: support@indiaaccounts.com
Web site: www.indiaaccounts.com



software is provided at a low cost, along with free upgrades. However, it charges a setup fee of Rs 750 and a monthly service fee of Rs 200 for home users and Rs 500 for small businesses.

All data is stored on the IndiaAccounts server. There is a 5 MB limit and the cost of buying additional space whenever needed.

SPECIFICATIONS

System Requirements: Windows 98/2000 or later
64 MB RAM, 500 MB free disk space, 56 Kbps Internet connectivity, and I.E 5.0 or higher

IndiaAccounts	
Performance	▶▶▶▶▶
Ease of Use	▶▶▶▶▶
Value for money	▶▶▶▶▶
Features	▶▶▶▶▶
OVERALL	▶▶▶▶▶

SiSoft Sandra 2003 Pro	
CPU Dryhstone	8071 MIPS
CPU Whetstone	3271 MFLOPS
CPU Integer	11923 instructions/sec
FPU SSE	12688 instructions/sec
ALU / RAM	2301 MB/s
FPU / RAM	2149 MB/s